EXP 4 – Different Prompting Technique

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GitHub Link: https://github.com/Sanjeev-123hub/Prompt-Engineering-EXP.NO-4.git

1.Comparative Analysis Prompt

Prompts:

1. I am going to buy a laptop or mobile for my college use, compare it and suggest me which one is better for me

2. Compare Google and Microsoft in terms of growth rate, annual income, etc.

Response:

1.

Feature	Mobile Phone	Laptop
Portability	Very easy to carry	Portable but needs a bag
Screen Size	Small screen	Large screen
Typing	Touch keyboard (slow for long typing)	Physical keyboard (best for assignments)
Software	Limited apps	Full software like VS Code, Excel, programming tools
Multitasking	Limited	Can run many apps together
Coding / Programming	Almost impossible	Perfect for coding and development
Assignments / Projects	Difficult	Easy to create documents, PPT, projects
Productivity	Mostly for communication	Best for studying and productivity

Laptops support **full software, better multitasking, and programming tools**, which are necessary for college projects and assignments.

2.

Factor	Google (Alphabet)	Microsoft
Founded	1998	1975
Founders	Larry Page & Sergey Brin	Bill Gates & Paul Allen
Headquarters	California, USA	Washington, USA
Annual Revenue (2025)	**~ $403 billion**	**~ $282 billion**
Revenue Growth Rate	**~ 15% yearly**	**~ 15% yearly**
Net Income	**~ $132 billion**	**~ $102 billion**
Operating Margin	**~ 32%**	**~ 44–47%**
Market Capitalization	**~ $3.7 trillion**	**~ $2.8 trillion**
Main Business Areas	Search, Ads, YouTube, Cloud, AI	Software, Cloud, AI, Office tools
Cloud Platform	Google Cloud	Microsoft Azure
Cloud Growth Rate	**~ 48%**	**~ 39%**

📈 Growth Drivers

Google

- Dominates **online search and digital advertising**.
- Major platforms: **Search, YouTube, Android, and Google Cloud**.
- Strong growth in **AI and cloud services**.

Microsoft

- Strong **enterprise software ecosystem**.
- Major products: **Windows, Office, Azure Cloud, LinkedIn, Xbox**.
- Rapid growth due to **AI integration and cloud computing (Azure)**.

2.Experiential Perspective Prompt

Prompt:

1. As a Machine Learning Expert, Explain What is ML?

2. As a Fullstack Developer, suggest me which one is better, MERN stack or MEAN stack which one is better?

Response:

1. What is Machine Learning?

Machine Learning (ML) is a branch of Artificial Intelligence that allows computers to **learn from data and improve their performance automatically without being explicitly programmed**.

In simple terms, instead of writing exact rules for a computer, we **give data and let the system learn patterns from it**.

Simple Definition

Machine Learning is the process of training a computer system using data so that it can make predictions or decisions automatically.

Example

Consider an **Email Spam Filter**.

1. The system is given **thousands of emails** labeled as *Spam* or *Not Spam*.
2. The ML algorithm studies patterns like:
    a. Certain keywords
    b. Sender information
    c. Email structure
3. After learning these patterns, the system can **automatically classify new emails as Spam or Not Spam**.

So the computer **learns from previous data instead of being manually programmed**.

2. 1. What is MERN Stack?

MERN = MongoDB + Express.js + React + Node.js

- **MongoDB** – NoSQL database
- **Express.js** – Backend framework
- **React** – Frontend library
- **Node.js** – Runtime environment

Key Idea: Uses **React** for building interactive user interfaces.

Advantages

- Very **popular in industry**
- Huge **community support**
- **Flexible architecture**
- Used by many startups
- Easier integration with modern tools

2. What is MEAN Stack?

MEAN = MongoDB + Express.js + Angular + Node.js

- **MongoDB**
- **Express.js**
- **Angular**
- **Node.js**

Key Idea: Uses **Angular**, which is a complete frontend framework.

Advantages

- Structured architecture
- Good for **large enterprise applications**
- Built-in tools and features

3. Key Differences

Feature	MERN Stack	MEAN Stack
Frontend	React	Angular
Learning curve	Easier	Slightly harder
Flexibility	Very flexible	More structured
Industry demand	Very high	Moderate
Community	Huge	Large but smaller than React

4. Which One is Better for You?

Since you already said:

- You **completed HTML & CSS**
- You **know basic JavaScript**
- You are learning **Full Stack Development**

MERN Stack is better for you.

Reasons

- **React is easier to learn than Angular**
- More **job opportunities**
- Used widely in **startups and modern web apps**
- Fits well with your **JavaScript learning path**

3. Everyday Functioning Prompts:

Prompt:

1. Suggest a healthy breakfast recipe for today.

2. Give me a 7th day task for me in prompt engineering learning

Response:

1. **Healthy Breakfast Recipe: Vegetable Oats**

Ingredients:

- 1 cup oats
- 1 cup water or milk
- 1/4 cup chopped carrots
- 1/4 cup peas
- 1 small onion (chopped)
- Salt to taste
- 1 teaspoon oil

Preparation:

1. Heat oil in a pan and sauté the chopped onion.
2. Add carrots and peas and cook for 2–3 minutes.
3. Add oats and water (or milk).
4. Add salt and cook for about 5 minutes until the oats become soft.
5. Serve hot.

Benefit:
This breakfast is rich in fiber, vitamins, and energy, which helps you stay active throughout the day.

2. **Day 7 Task – Role-Based Prompting**

Objective

Learn how to **assign a role to the AI** so that it gives **more accurate and expert-level responses**.

Task 1: Create a Role Prompt

Write a prompt where you assign a professional role to the AI.

Example Prompt:

“Act as a fitness trainer and suggest a 10-minute morning exercise routine for beginners.”

Task 2: Try a Different Role

Use another role and ask a question.

Example Prompt:

“Act as a financial advisor and explain how a college student can start saving money.”

Task 3: Compare the Response

1. Ask the question **without a role**.
2. Ask the **same question with a role**.
3. Compare the answers and observe:
    a. Detail level
    b. Accuracy
    c. Structure of the response